MIL-T-62255C(AT)
31 October 1980
SUPERSEDING
MIL-T-62255B(AT)
6 March 1980

MILITARY SPECIFICATION

TRUCK TRACTOR: YARD TYPE, 4X2, M878

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel engine driven, four-wheel, two rear wheel drive truck-tractor, yard type, with a one man half cab, having a minimum gross vehicle weight rating (GVWR) of 43,500 pounds and a gross combination weight rating (GCWR) of 100,000 pounds. Vehicles under this specification shall be procured only from a manufacturer who meets the standard commercial and commercially proven component criteria. Vehicles shall be warranted by the manufacturer as specified in the procurement documents.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

ZZ-T-381	- Tires, Pneumatic, Vehicu1ar (Highway).
GG-P-455	- Plates and Foils, Photographic (Photo-sensitive Anodized Aluminum).
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.l, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2320

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TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking Blank.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordinance.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-K-62298	- Kit, Winterization: Trucks, Commercial.

STANDARDS

MILITARY

MIL-STD-193	- Finishing Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1223	- Administrative Wheeled Vehicles Treatment, Painting Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35000	- Batteries.
MS51117	- Pintle Assembly, Towing: 100,000 lbs. Capacity, Manual Release.
MS75004	- Terminal Lug (Storage Battery).
MS75020	- Connector, Plug, Electrical-12 Contact, Intervehicular, 28-volt, Waterproof.
MS75021	- Connector, Receptacle, Electrical-12 Contact, Intervehicular, 28-volt, Waterproof.

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DRAWINGS

7728811	- Harness Assembly, Intervehicular.
7731428	- Cover Assembly, Receptacle.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION
Federal Motor Carrier Safety Regulations
Federal Motor Vehicle Safety Standards

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment, Part III, Chapter I, Part 205, Title 40.

(Application for copies should reference "Code of Federal Regulations 40 CFR and the Federal Register" and should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

SAE J198	- Windshield Wiper Systems - Trucks, Buses and Multipurpose Vehicles.
SAE J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers-Trucks, Tractors and Trailers.
SAE J381	- Windshield Defrosting Systems Test Procedure - Trucks, Buses and Multipurpose Vehicles.
SAE J382	- Windshield Defrosting Systems Performance Requirements - Trucks, Buses and Multipurpose Vehicles.

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SAE J429	- Mechanical and Material Requirements for External Threaded Fasteners.
SAE J536	- Hose Clamps.
SAE J551	- Measurement of Electromagnetic Radiation from a Motor Vehicle or Other Internal Combustion Powered Device (Excluding Aircraft). (20-1000 MHz).
SAE J552	- External Electromagnetic Radiation Suppressors.
SAE J555	- Truck, Truck-Tractor, Trailer and Motor Coach Wiring.
SAE J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
SAE J598	- Sealed Lighting Units for Construction Industrial Machinery.
SAE J682	- Rear Wheel Splash and Stone Throw Protection, Commercial Vehicles.
SAE J688	- Truck Ability Prediction Procedure.
SAE J695	- Turning Ability and Off Tracking - Motor Vehicles.
SAE J700	- Fifth Wheel Kingpin.
SAE J704	- Side Openings for Truck Power Take-Offs.
SAE J816	- Engine Test Code - Spark Ignition and Diesel.
SAE J849	- Connection and Accessory Locations for Towing Double Trailers and Multi-Axle Trailers.
SAE J1067	- Seven Conductor Jacketed Cable for Truck - Trailer Connections.
SAE J1402	- Automotive Air Brake Hose and Hose Assemblies.

(Application for copies of the SAE Standards and Recommended Practices should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, Pennsylvania 15096.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.
Metric Practice
Z-210. Metric Practice (ASTM E 380).

(Application for copies should be addressed to American National Standards Institute Incorporated, 1430 Broadway, New York City, New York 10018.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

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(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, Ohio 44313.)

STATE OF CALIFORNIA
California Vehicle Code

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TEST OPERATION PROCEDURES
TOP 2-2-614 Toxic and Hazard Test for Vehicles and other Equipment

(Application for copies should be addressed to Commander, U.S. Army Test and Evaluation Command, ATTN: DRSTE-ME, Aberdeen Proving Ground, Maryland 21005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Commercial truck-tractor, half-cab. The commercial truck-tractor, half-cab will be a vehicle which is the manufacturer's standard model with standard optional items conforming to all requirements of the solicitation and has been manufactured on a production basis, and sold to the public for at least one calendar year immediately preceding the date of issuance of this solicitation. Unless otherwise specified in procurement documents, chassis items shall be as represented and rated in the manufacturer's technical data, and mounted equipment shall be as represented and rated in the equipment manufacturer's technical data. Technical data shall consist of the specification and technical material furnished to authorized company representatives for selection of vehicle models and components.

3.1.1 Special requirements. In addition to standard vehicle and components specified in 3.1, vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Treatment, painting, marking, data plates and warranty notice.

3.1.1.1.1 Treatment and painting. Unless otherwise specified by the procuring activity for the appropriate service (see 6.2), vehicles shall be treated and painted in accordance with MIL-STD-193 for the Army.

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a. Exterior metal and plastic surfaces. All exterior metal and plastic surfaces shall be treated as specified in MIL-STD-193 and finish painted with an acrylic or an alkyd base paint having color, gloss and infrared reflectance properties of lusterless forest green in accordance with MIL-E-52798 or MIL-E-52835. Metal surfaces shall be pretreated as specified in Type I or Type III of TT-C-490. Surfaces shall be primed as specified in TT-P-636. When overcoated with acrylic or alkyd enamel in accordance with the requirements of MIL-E-52798 or MIL-E-52835, there shall be sufficient intercoat adhesion to assure a good paint bond.

b. Chassis, accessories and components. The chassis rear of the vehicle cab, rear axle, fifth wheel lifting arms, tread plate, steps, floodlamps, handles, fenders, pintle, fuel tank, air and hydraulic tanks, exterior of cylinders and all components shall be finish painted in accordance with MIL-E-52798 or MIL-E-52835.

c. Exterior model name plates and trim. Exterior model name plates and ornamental trim may be eliminated, providing no holes are left. Small external screws that are normally bright finished may be black phosphated or chromated, in lieu of the lusterless forest green paint specified herein. Stainless steel lock cylinders and flat-finished windshield wiper arms, blades and pivots need not be painted.

d. Manufacturer's standard painting procedure. Top surface of the fifth wheel assembly; top surface of the fifth wheel fork and approach ramps; front under body components, engine and chassis forward of the rear vehicle cab, and front underbody components not visible from 5 feet from the vehicle may be treated in accordance with manufacturer's standard commercial practice.

e. Rubber and seals. Rubber, gaskets and seals need not be painted.

f. Interior plastic and metal surfaces. The interior plastic accessories and metal surfaces of the cab shall conform to manufacturer's standard commercial practice.

g. Vehicle cab interior. The vehicle cab interior side lining, ceiling, header and upholstery color shall be manufacturer's standard black, dark green or dark brown. All vehicles shall have the same interior color and upholstery color.

3.1.1.1.2 Marking The markings shall be in accordance with MIL-STD-642 except as specified herein. Exterior markings shall be black lusterless paint having equivalent infrared

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reflectance of MIL-E-52798 or MIL-E-52835, and informational markings shall be located as follows:

(a) Safety and instructional markings. Markings such as tire pressure, fuel type, fill level of liquids, towing, lifting, tiedown and center of gravity shall be black lusterless letters or numerals no larger than one inch.

(b) Concealed marking. When specified (see 6.2), concealed marking shall consist of the registration number only, and shall be applied as specified in 4.3 of MIL-STD-642.

(c) Vehicles supplied to foreign nations. When specified (see 6.2), vehicles supplied to foreign nations, temporary registration numbers shall be furnished as specified in 4.4 of MIL-STD-642.

3.1.1.1.3 Data plates and warranty notice. Instruction, caution, identification, military model designator and National Stock Number (NSN), operating and data plates shall be furnished and installed. The plates shall be in accordance with Type III, Composition A or Composition C of MIL-P-514, or standard commercial practice. When foil is employed for Composition C, the plate shall be in accordance with Type I, Class I of GG-P-455. Foil shall not be used for exterior plates. Vehicle data plates and warranty notice shall be in accordance with MIL-STD-1223.

3.1.1.2 Rustproofing. Vehicle shall be rustproofed in accordance with MIL-STD-1223. Undercoating is not acceptable. In addition, all chassis frame members, enclosed areas, boxed-in areas, such as lights and under floor pan, and the vehicle cab shall be rustproofed. The material may be applied by using an atomized spray, pressure gun or brush. Metal surfaces shall be cleaned prior to the application of rustproofing. Dissimilar metals shall be installed as specified in 5.8 and Table IX of MIL-STD-193. Certified test report shall be furnished to the engineering office of the procuring activity.

3.1.1.3 Heavy duty cooling system. Heavy duty cooling system shall be furnished. The cooling system shall maintain the coolant at a temperature below the boiling point, with the vehicle loaded to the rated GVWR and operated at sea level, and in an ambient air temperature of not less than 125°F. Cooling system radiator shall be of at least 1400 square inches and of sufficient depth to meet the cooling requirements specified. Certify that the cooling system is matched and rated in conjunction with interfacing components.

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3.1.1.3.1 Coolant system and indicators. Coolant system shall inciude a deaeration system and a surge tank or a coolant recovery reservoir of not less than two quart capacity. A high coolant temperature and a low coolant level alarm buzzer and red indicator warning light shall be provided on the dash instrument panel. The reservoir shall be located in a position readily visible for inspection of the coolant level. A radiator servicing access door shall be provided.

3.1.1.4 Towing, tiedown and lifting devices. Towing, tiedown and lifting devices specified herein, shall be furnished. The tiedown and lifting devices shall be in accordance with Class 1, Type II of MIL-STD-209, and their attaching members shall be certified to the engineering office of the procuring activity that the devices meet or exceed MIL-STD-209.

3.1.1.4.1 Towing devices. Two towing devices shall be furnished on the front of the vehicle and two towing devices shall be furnished on the rear of the vehicle. The towing devices shall be capable of towing the specified GCWR. A caution plate shall be provided depicting the following.

NOTE: Do not tow vehicle over 20 miles per hour (mph) and no more than 15 miles.

3.1.1.4.2 Tiedown and lifting devices. In addition to the towing devices specified in 3.1.1.4.1, the vehicle shall be capable of being transported by highway, rail, ocean carrier and C5A aircraft. Complete diagrams and instructions for lifting and tying down the vehicle to the various transportation modes shall be provided. The tiedown devices shall permit tiedown of the vehicle to the floor or deck of the transportation medium in such a manner as to prevent shifting or movement in any direction. Each device and its attaching member shall be capable of accepting its proportionate share of the loads imposed.

3.1.1.5 Rear crossmember mounting for pintle assembly. A rear crossmember for mounting a pintle assembly MS51117 by the receiving activity shall be furnished. Location for the holes for the pintle assembly shall be as specified in 3.1.1.5.1. The crossmember shall include the mounting holes for the pintle assembly, and the two towing devices specified in 3.1.1.5.1. The brake air lines and the electrical lines shall terminate forward of the crossmember with provisions for the receiving activity to install the air brake receptacles and electrical 12 volt

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receptacle. The crossmember shall be inscribed for the location of the air and electrical connectors. The crossmember shall be capable of accepting the loads imposed and the mounting depicted in SAE J849. Instruction shall be provided in the manual for the vehicle. A drawbar test to verify that the crossmember shall meet or exceed a pull of 60,000 pounds shall be conducted.

3.1.1.5.1 Rear-mounted pintle assembly. When specified (see 6.2), a rear-mounted pintle assembly shall be furnished. The pintle assembly shall be of the swivel type and conform to MS51117. The assembly shall be furnished with mounting flanges and lubrication fittings. The pintle assembly shall be mounted forward, but not more than four inches forward, of the rearmost part of the vehicle. The pintle assembly shall be mounted 34 inches above ground level, plus or minus one inch, measured at the centerline of the pintle shaft, to provide a lunette height on a dolly drawbar of 32 to 36 inches above ground level. Mounting shall include reinforcements to transfer pintle loads directly to the web of the chassis frame. Pintle assembly shall be mounted so as not to damage the drawbar of the dolly and shall be capable of assuming an angle of not less than 70 degrees with the horizontal axis of the vehicle without damage to the vehicle. Air connectors for a dolly or full trailer brake, and the lighting receptacles conforming to SAE J560 with cover assembly, shall be mounted in a readily accessible and protected location near the rear pintle. The two towing devices specified in 3.1.1.4.1 shall be employed for use of safety chains that are secured to the dolly or full trailer drawbar. Receptacles and air connectors shall be located as specified in SAE J849. Pintle location shall be as specified herein.

3.1.1.1.6 Electromagnetic interference (EMI). When specified (see 6.2), electromagnetic interference suppression shall be furnished. The vehicle shall be suppressed to the level specified in paragraph 4.1.2, Notice 4 of MIL-STD-461, Radiated Emission Requirement REO 5, at a distance of two meters referenced in Table VIII, with limits shown in Figures 13 and 18, and Conducted Emission Requirement CEO 7, limit of Figure A-4. In addition, when arctic provisions specified in 3.1.1.10 are furnished, the operation of the installed arctic provisions shall not cause the overall vehicle electromagnetic interference characteristics to fail to meet paragraph 4.1.2 of Notice 4 of MIL-STD-461. The vehicle air horn, turn signals and starter are excluded.

3.1.1.6.1 Radio interference suppression. Unless otherwise specified in 3.1.1.6, the vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Suppressors used shall be in accordance with SAE J552.

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3.1.1.7 Drain plugs. Drain plugs installed in the transmission, power takeoff, axle differential and the hydraulic fluid reservoir, shall be of the permanent magnet type.

3.1.1.8 Emergency equipment. Emergency equipment shall be furnished. The emergency equipment shall consist of reflective triangles in accordance with Federal Motor Vehicle Safety Standards 125 and Federal Motor Carrier Regulation 393.95. Fire extinguisher and tire chains shall not be furnished. Emergency equipment shall be stowed in a metal container in the vehicle cab.

3.1.1.9 Splash shields. Removable rubber splash shields shall be furnished for the rear wheels. The splash shields shall be mounted in accordance with SAE J682. Splash shields shall be mounted to the vehicle to preclude the splash shield from rolling over the top of the tire when the vehicle is in reverse, and shall be not below the top of rear axle.

3.1.1.10 Arctic provisions. When specified (see 6.2), arctic provisions shall be furnished in accordance with MIL-K-62298. The arctic provisions shall be employed when the vehicle is required to meet Climatic Category 7 as specified in 3.2.4.2 and Table I.

3.1.1.11 Carbon monoxide and NO_X concentration. The vehicle and heater exhaust system shall be installed in a manner that will insure that the concentration of NO_X shall not exceed 5 parts per million (ppm), and the concentration of carbon monoxide shall not exceed 50 ppm (0.0050) in the drivers compartment, after eight hours of operation with all heaters operating and the vehicle engine at idle speed. Certified test report shall be provided to the engineering office of the procuring activity. Carbon monoxide and NO_X may be tested in accordance with TOP 2-2-614, Toxic Hazards Test for Vehicles and other Equipment. Contractor shall indicate levels obtained and test equipment used.

3.1.1.12 Flame resistance. Flame resistance shall be in accordance with Federal Motor Carrier Safety Regulation 393.84 and Federal Motor Vehicle Safety Standard 302. Certified test report shall be provided to the engineering office of the procuring activity.

3.1.1.13 Hydraulic fluid. Hydraulic fluid shall be furnished. The hydraulic fifth wheel hydraulic system fluid shall be in accordance with MIL-H-5606 or commercial equivalent to meet or exceed the climatic conditions in which the vehicle will operate.

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3.1.1.14 Lubricants and fuel. Military equivalent of commercial recommended lubricants and fuel shall be depicted on a plate (see 3.1.1.1.3) and installed in a readily accessible and visible location in the vehicle cab. The plate shall indicate ambient temperature range for the various lubricants and fuel.

3.1.1.15 First aid kit. First aid kit shall be furnished and installed in the vehicle cab in an accessible location. First aid kit shall conform to Federal Motor Carrier Safety Regulation 393.96, Type B, Commercial Type Kit. The first aid kit shall be constructed of aluminum and as specified in 393.86 (1) through 393.86 (7).

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standars. Vehicle and furnished accessories and components shall comply with all applicable Federal Motor Vehicle Safety Standards in effect on the date of manufacture, except as specified herein.

3.2.2 Air pollution control. Vehicle shall comply with the Environmental Protection Agency Regulations governing control of air pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles shall comply with the State of California Regulations governing air pollution control in effect on the date of manufacture as applicable. Vehicle shall be labeled.

3.2.3 Sound level. The vehicle shall be labeled indicating that the interior and exterior noise level meets the requirements specified in 3.2.3.1 and 3.2.3.2.

3.2.3.1 Interior sound level. Interior sound level shall be measured in accordance with Table I, Category D of MIL-STD-1474, and shall be less than 85 db (A). Interior noise level shall be tested with the windows, vents and door in the open position and in the closed position, as stated in 5.1.2.1.5 of MIL-STD-1474.

3.2.3.2 Exterior sound level. Exterior sound level shall be in accordance with the Environmental Protection Agency Noise Emission Standards for Transportation Industry, Part III, Title 40, Chapter I, Part 205.

3.2.4 Vehicle components subjected to climatic environment. Vehicle components and material that will be subjected to climatic environment shall be capable of accepting the requirements specified in Table I without evidence of deformation, cracking or unusual wear, and shall not affect the vehicle warranty as expressed in procurement documents.

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3.2.4.1 Climatic conditions. The vehicle shall be capable of operation in ambient temperatures from +125°F to -25°F, as specified in categories 1 through 6 of Table I.

3.2.4.2 Climatic conditions, extreme. When specified (see 3.1.1.10), the vehicle shall be capable of operation in ambient temperatures of -25°F to -50°F, as specified in Category 7 of Table I.

3.2.4.3 Climatic conditions, storage and transit conditions. Vehicle shall be capable of meeting the storage and transit conditions as specified in Categories 1 through 7 of Table I without evidence or malfunction on any component of the vehicle.

TABLE I. SUMMARY OF TEMPERATURE, SOLAR RADIATION, AND RELATIVE HUMIDITY DIURANAL EXTREMES

	OPERATIONAL CONDITIONS			STORAGE AND TRANSIT CONDITIONS	
CLIMATIC CATEGORY	AMBIENT AIR TEMPERATURE-DEGREES F	SOLAR RADIATION-BTU/FT2/HR	AMBIENT RELATIVE HUMIDITY-%	INDUCED AIR TEMPERATURE DEGREES F	INDUCED RELATIVE HUMIDITY %
1 WET-WARM	Nearly Constant 75	Negligible	95 to 100	Nearly Constant 80	95 to 100
2 WET-HOT	70 to 95	0 to 360	74 to 100	90 to 160	10 to 85
3 HUMID-HOT COASTAL DESERT	85 to 100	0 to 360	63 to 90	90 to 160	10 to 85
4 HOT-DRY	90 to 125	0 to 360	5 to 20	90 to 160	2 to 50
5 INTER-MEDIATE HOT-DRY	70 to 110	0 to 360	20 to 85	70 to 145	5 to 50
6 INTER-MEDIATE	-5 to -25	Negligible	Tending Toward Saturation	-10 to -30	Tending Toward Saturation
7 COLD	-25 to -50	Negligible	Tending Toward Saturation	-35 to -50	Tending Toward Saturation

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3.2.5 Curb weight. Curb weight of the vehicle shall include the weight of the chassis and cab, with all attachments, accessories and equipment; fifth wheel; full complement of fuel lubricants and coolant.

3.2.6 Gross vehicle weight rating. Gross vehicle weight rating (GVWR) shall consist of the curb weight, operator (weight computed at 200 pounds), and a 28,000 pound load imposed on the fifth wheel. The load imposed on the fifth wheel shall be so distributed that the vehicle GVWR rating and the design capacities of the chassis and components are not exceeded.

3.2.7 Gross combination weight rating. Gross combination weight rating (GCWR) shall consist of the curb weight, operator (weight computed at 200 pounds), and weight of a semitrailer loaded to provide not more than 100,000 pound GCWR.

3.2.8 Ratings. Vehicle ratings shall be manufacturer's published ratings for Class I road as defined in Table 9 of SAE J688. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted. Minimum GVWR and GCWR shall conform to 3.2.6 and 3.2.7.

3.2.9 Dimensions. Dimensions for the vehicle in the unladen position shall not be more than that specified in Table II.

TABLE II. DIMENSIONS

	Inches
Fifth wheel height:	
a. Down position from ground level (plus 1 inch - minus 0)	48
b. Raised position from ground level (minimum)	64
c. When specified (see 3.5.5), raised position at least	79
Landing Wheel clearance (maximum).	57
Centerline of axle to rear of frame (maximum).	21 1/2
Swing radius, not including fifth wheel travel forward when being raised, not less than	68
Ramp height at rear of chassis (plus or minus 1 inch)	36

NOTE: The towed vehicle is a 40-foot semitrailer for transporting one 40-foot or two 20-foot containers. Dimensions are as follows:

Overall length	40	feet
Kingpin location from front	30	inches
Overall height, platform to ground	58	inches
Overall width	96	inches
Overall height with payload	162	inches max.
Center of gravity rearward of kingpin	207	inches
Center of gravity above ground level	109	inches
Tri-axle spacing	50	inches
Semitrailer gross weight	82,000	pounds

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3.2.10 Materials. Material shall be as specified herein. All material shall be new and unused. Materials not specified shall be selected by the supplier as being compatible with the intended use in accordance with the engineering standards of the heavy truck and construction equipment industry and as set forth in the following industry standardization publications:

Society of Automotive Engineers (SAE)
Tire and Rim Association (TRA)
American National Standards Institute (ANSI)
Industrial Fasteners Institute (IFI)
Truck Trailer Manufacturer's Association (TTMA)
American Society of Testing and Materials (ASTM)
Friction Materials Standards Institute (FMS)

3.2.11 Metric components in equipment. The use of metric components in equipment shall be identified. The contractor shall identify metric tool requirements for each level of maintenance. Metric usage shall conform to ASTM-E-380, Metric Practice Guide (ANSI-Z-210). Metric equivalents may be used either as the primary standard of measurement, or as the secondary standard, in parentheses. Where metric equivalents in parentheses are not considered practicable, a tabulation of equivalents shall be depicted.

3.2.12 Screw threads and fittings. Screw threads and fittings shall be based on United States Standards and shall be identified when metric components are used.

3.2.13 Recycled, virgin and reclaimed materials. The use of recycled, virgin and reclaimed materials which meets or exceed the requirements of the applicable material specifications without jeopardizing the intended use of the end item shall be encouraged.

3.2.14 Fuel conservation. Engine and power train combinations selected shall be the most fuel efficient components which otherwise meet specified performance criteria. Engineering data and performance calculations submitted under requirements of the solicitation shall include fuel economy calculations based on a road speed of 30 miles per hour (mph) Class 1 Roads, Factor-Fair as specified in table 9 of SAE J688, at a gross combination weight (GCW) of the vehicle combination, with a gradeability of 0.5 percent, ambient temperature of approximately 60°F. In addition, fuel consumption shall be provided with the vehicle at idle speed. The coolant temperature of the engine and the transmission and converter shall be provided.

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3.3 Performance. Vehicle will operate primarily in shipping terminals, shuttling transport semitrailers from ship side to marshalling yards and trailer transfer points. Vehicle shall be capable of operating on Class I roads in accordance with TABLE IX of SAE J688 and grades and speeds specified herein.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the truck tractor loaded to specified GCWR.

3.3.1.1 Gradeability. Vehicle shall be capable of ascending a continuous grade of 1.0 percent at 20 miles per hour (mph). Gradeability shall be verified with calculations in accordance with SAE J688, Truck Ability Prediction Procedure (see 6.3). Gradeability in low gear shall be not less than 27 percent. Calculations shall indicate:

a. Ratio, revolutions per minute and miles per hour in each range.
b. Torque, horsepower and revolutions per minute and calculated grade that the GCW will be capable of ascending in each of the forward gear ranges.
c. Cooling system calculations to verify cooling requirements at minus 25°F ambient temperature and the grades that the vehicle will be subjected to as specified in 3.1.1.3 and 3.3 as specified herein.

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of the recommended governed speed, and shal1 provide not more than three mph.

3.3.1.3 Reverse speed. The vehicle shall be capable of a reverse speed of five to ten mph.

3.3.1.4 Maximum geared speed. Maximum geared speed at engine governed speed shall be 35 mph, plus or minus 15 percent. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x (tire factor)}}$$

3.3.1.5 Inching capabilities. Inching capabilities shall be demonstrated from a smooth forward or reverse starting position.

3.3.1.6 Sustained operation. The vehicle shall be capable of sustained operation for 10 to 12 hours without scheduled maintenance or servicing.

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3.3.2 Service brakes. Service brakes stopping distance shall be in accordance with the requirements of Federal Motor Carrier Safety Regulation 393.52. The anti-skid requirements of Federal Motor Vehicle Safety Standard 121 shall not be furnished or its components and accessories installed on the vehicle. A means shall be furnished and installed to decrease the braking forces on the front wheel brakes when the anti-skid requirements are not furnished.

3.3.2.1 Parking brake. Parking brake shall be furnished. Parking brake shall be in accordance with Federal Motor Carrier Safety Regulation 393.41.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial vehicle which meets or exceeds the requirements of this specification. Vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel turbocharged engine, with not less than six cylinders. Engine net horsepower figures used in the performance prediction calculations shall be determined in accordance with SAE J816. Maximum net engine torque shall be not less than 550 pound-feet at 1800 revolutions per minute, and develop not less than 225 gross horsepower at 2,600 revolutions per minute (rpm) and provide an engine torque of 455 pound-feet at 2,600 rpm. The engine shall not display any oil downstream of the exhaust system when engine is idling at manufacturer's recommended idle speed. Fuel consumption shall be observed in the selection of the engine and power train requirements in accordance with 3.2.14. Certify that the engine is matched and rated for use in conjunction with interfacing components.

3.4.1.1 Filter, coolant system. A filter for the coolant system shall be furnished. The filter shall be of the spin-off type for filtering of the coolant contaminates.

3.4.1.1.1 Fan clutch. A thermostatically-controlled fan clutch shall be furnished and installed for the cooling system. Vehicle performance shall be with the fan clutch and fan engaged. Fan clutch shall have a positive lock-up in case of failure of the air control system. Fan bolts shall be Grade 8 in accordance with SAE J429. Bolts shall be torqued. Fan and fan clutch assembly shall be of aluminum to decrease stress on the fan shaft.

3.4.1.1.2 Fan shroud. Fan shroud shall be furnished. The fan shroud shall be around the perimeter of the engine fan and shall be securely fastened to preclude vibration of any component, and shall be capable of removal for service of front engine components.

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3.4.1.1.3 Water pump. Engine water pump, with full blocking thermostats at engine governed speed, shall be furnished.

3.4.1.1.4 Coolant hoses and hose clamp. Coolant hoses shall be of the silicone rubber type and shall be capable of accepting a minus 65°F ambient temperature. Hoses shall be secured with double wrap stainless steel; worm type gear hose clamps in accordance with Type F, class 4 of SAE J536.

3.4.1.1.4.1 Coolant temperature. The coolant temperature of the engine shall be the engine manufacturer's recommended operating temperature and shall be not less than 175°F.

3.4.1.2 Oil filters. A full flow, spin-off type oil filter and a by-pass type oil filter shall be furnished. Oil filters shall be in a protective location from road elements and low ambient temperatures.

3.4.1.2.1 Oil cooler. An oil cooler shall be furnished. The oil cooler shall be provided with a water drain. The oil cooler shall be of sufficient capacity to meet the speed, ambient temperature and gradeability requirements to which the vehicle will be subjected.

3.4.1.3 Fuel system. Fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.1.3.1 Fuel pump. Manufacturer's commercial positive displacement, gear-type fuel pump shall be furnished.

3.4.1.3.2 Fuel tank(s). Fuel tank(s) shall have a total capacity of not less than 50 gallons. Guard on the exterior of the fuel tank(s) shall be furnished. Guard shall be bolted to enable service. Guard to protect fuel tank while working in a close environment. The fuel tank and the fuel system shall be capable of providing the required amount of fuel to the vehicle engine when ascending and descending a grade of 30 percent and side slopes of 10 percent. A red warning indicator lamp shall be provided on the instrument panel to indicate fuel in fuel tank is too low for the slope and grade requirements specified herein. Fuel lines shall be protected from build-up of ice, snow, road splash and stone throw.

3.4.1.3.3 Air cleaner. A heavy duty type air cleaner with a service indicator shall be furnished as recommended by the engine manufacturer for operation under severe dust conditions.

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3.4.1.3.4 Air induction system. An engine air induction system shall be furnished and installed from the air cleaner to the engine intake. Air cleaner indicator shall be provided that is visible to the driver. Means shall be provided to preclude water entering the air system. When the air cleaner is stacked and is above the vehicle cab, a means shall be provided to either lower or swing the assembly down to enable the vehicle to enter buildings or for transport or storage. The air cleaner shall be designed to preclude the entrance of blowing snow.

3.4.1.3.5 Governor. Engine governor shall be furnished and set and sealed to limit engine to engine manufacturer's maximum recommended operating speed. Governor foot throttle control linkage shall be adjusted to permit the full range of the engine governed speed. The accelerator controls system shall conform to Federal Motor Vehicle Safety Standard 124.

3.4.1.3.6 Engine starting. Engine starting equipment shall include an ether or glow plug starting system. The ether starting system shall be of the automatic, ignition, or engine temperature limited type, controlled from the driver's area and be inoperable with the engine operating. When glow plugs are furnished in lieu of the ether starting system, they shall be provided for each cylinder. Engine starting equipment shall start the engine in an ambient temperature of minus 25°F. Appropriate type of lubrication for the ambient temperature that the vehicle is operating shall be provided.

3.4.1.3.7 Power plant heaters. Power plant heaters shall be provided for the coolant, engine oil and the batteries. Heaters shall operate on 110-volt ac, and shall be wired through a junction block to a single three pronged male, weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard, three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation, of connecting cable, shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

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(a) Coolant heaters, 1500-watt minimum rating, shall be installed in the engine block or lower, coolant, inlet hose. A coolant circulating pump, driven by a 110-volt ac motor, shall be provided. Engine thermostat with an operating range of 170°F to 195°F shall be installed.
(b) Immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in oil pan through any convenient opening.
(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperature as low as 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

3.4.1.3.8 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. At least the muffler(s) tailpipe(s), raincap(s) and heat shield(s), and associated hardware shall be corrosion resistant. The system shall be secured and braced to resist vibration. The tail pipe outlet shall be located to prevent exhaust fumes entering the vehicle cab. Any portions of the system which project rear of the cab shall be provided with a cover plate. The cover plate shall form a shield that will shed fluids coming in contact with the system. The system shall discharge to vertical stack(s). The vertical portion and any component that personnel will come in contact with shall be furnished with a heat shield. The tail pipe shall be provided with a means of either lowering the tail pipe or swinging the tail pipe below the cab roof line when entering buildings or for transport or storage. The system shall be tight to preclude any accumulation of carbon monoxide in occupied areas and on personnel servicing the intervehicular cables.

3.4.2 Electrical (climatic extremes). The electrical wiring, components and accessories shall be capable of performing their function without excessive voltage drop when operating in Climatic Category 7 specified in Table I.

3.4.2.1 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33. The vehicle will be operating at sea ports. The electrical system shall not be affected by sea breeze or blowing sea mist.

3.4.2.1.1 Wiring. Wiring shall conform to SAE J555. All wiring rear of the vehicle cab shall be fully insulated, supported and protected from chafing. Wiring shall be weatherproof. Wiring shall be color and number coded for identification. Wiring shall be so positioned that

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there shall not be any interference with any vehicle component that will result in chafing or failure. Wiring shall be harnessed.

3.4.2.2 Starting system. The starting system shall be a 12-volt starting system.

3.4.2.2.1 Cranking motor solenoid. The cranking motor solenoid shall be accessible for service and protected from road splash and meet the climatic requirements specified herein.

3.4.2.2.2 Cranking motor circuit. The cranking motor circuit voltage drop shall be not greater than 0.20 volts per 100 amperes of the cranking motor circuit.

3.4.2.3 Batteries. A minimum of four 12-volt batteries shall be furnished. Batteries shall be in accordance with type 6 TN of MS35000. Battery terminal lugs shall be in accordance with MS75004. The ground wire to the frame shall be bolted in lieu of welding to the vehicle frame.

3.4.2.3.1 Battery compartment. Battery compartment, tray, brackets and cover shall be furnished. Compartment shall be of sufficient structural strength for accepting the imposed loads during travel, as specified herein. The tray, brackets, compartment interior and inside of the compartment cover shall be treated and painted with acid resistant paint in accordance with TT-C-494. Battery hold-down brackets shall be furnished. Compartment shall be vented and interior protected from road splash. The area between the top of the battery posts and the underside of the cover shall be provided with clearance. To preclude condensation dripping on the batteries, the battery box may be tilted rearward and outward or the underside of the cover may be crowned. Compartment shall be provided with a hasp for a 1 1/2 inch padlock. For shipping purposes, cover shall be secured to preclude theft when in transit.

3.4.2.3.1.1 Battery hold down brackets. Each battery shall be provided with hold down brackets. Brackets shall enclose each battery case. Brackets shall be treated to resist corrosion from batteries. Each battery shall be placed on a heavy, thick battery blanket. When vehicle is traveling, vibration shall not cause the battery(s) to crack.

3.4.2.3.2 Alternator. An alternator shall be furnished. The alternator shall provide for 12 volt lighting and starting capabilities and provide not less than 85 ampere hour capacity. Alternator shall be grounded by bolt and to the vehicle frame.

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3.4.2.3.3 Sheaves. Sheaves shall be the correct size in width and length in accordance with the pitch diameter of the sheaves. Fan, alternator, power steering pump and air compressor, if belt driven, shall provide sufficient clearance between belts without interference of any component. Sheaves for the fan assembly shall be as close to the fan bracket belt adjustment as possible to decrease stresses on the bracket and hub.

3.4.2.4 Lighting. Lighting shall be 12-volt. Lighting, reflectors and wiring shall conform to FMVSS 108. Rear lighting shall be mounted in a protective location to preclude damage when interfacing with towed vehicles or body operations. Combination lighting on cab roof shall be controlled by an on and off switch in the vehicle cab and dash mounted for tactical operations.

3.4.2.4.1 Turn signals. Turn signals shall conform to FMVSS 108. Operating mechanism shall be self-cancelling type.

3.4.2.4.2 Floodlights. Two 12-volt floodlights shall be furnished. Floodlight shall be installed at the rear of the vehicle cab on U brackets. Floodlight shall be sealed-beam, adjusted vertically on the U-frame. Floodlight exterior shell shall be finished painted as specified in 3.1.1.1. Light shall conform to SAE J598, Trade No. 4478. Light shall be provided with an on and off heavy duty metal switch in the vehicle cab dash mounted. Light and light switch shall be weatherproof. The lights shall be capable of illuminating container numbers as vehicle drives by containers that are stacked in aisles.

3.4.2.4.2.1 Panel-mounted floodlamp. A panel-mounted floodlamp shall be furnished. The floodlamp shall be mounted on a vertical surface at the rear of the vehicle cab. Floodlamp shall be operated by a metal, heavy duty switch, operable from the vehicle dash panel. Lamp shall focus on the fifth wheel. Floodlamp shall be waterproof. Lens shall be unbreakable. Exterior shall be finished painted as specified in 3.1.1.1.

3.4.2.5 Lighting receptacles, 12-volt.

3.4.2.5.1 Lighting receptacle, 12-volt (back of cab). A 12-volt lighting receptacle conforming to SAE J560, with its connectors connected and color coded as specified herein, shall be furnished. The receptacle shall be furnished complete with a spring-loaded cover.

3.4.2.5.2 Lighting receptacle, 12-volt (rear of vehicle). When a rear pintle assembly is specified in 3.1.1.5.1, a 12-volt receptacle, as specified in 3.4.2.5.1, shall be furnished in a

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protective and accessible location. When a crossmember for a pintle assembly is to be assembled by the receiving activity, the wiring and location shall be specified in 3.1.1.5.

3.4.2.5.3 Lighting cable, 12-volt. A 12-volt coiled retractable lighting cable conforming to SAE J1067 shall be furnished. The coiled lighting cable shall be not less than 10 feet in 1ength and both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a spring cable guard on both ends and a grip for withdrawing from the receptacles. The end of the cable shall be furnished with a spring-loaded cover to preclude contaminates entering the socket.

3.4.2.5.4 Lighting for 28-volt towed semitrailers. A 28-volt lighting system shall be furnished (see 3.4.2.5.4.3).

3.4.2.5.4.1 Lighting receptacle, 28-volt (rear of cab). A 28-volt lighting receptacle conforming to MS75021-2 with cover assembly conforming to Drawing 7731428, shall be furnished. Receptacle shall provide lighting when towing a semitrailer with 28-volt system. The 28-volt, 12 ampere output service shall be connected to terminals B, D, E, J and L of MS75021-2.

3.4.2.5.4.2 Lighting cable, 28-volt. A 28-volt conductor lighting cable conforming to Drawing 7728811 shall be furnished. Cable shall be not less than 10 feet in length and both ends shall be equipped with connectors conforming to MS75020-1 and MS75020-2. Cable may be of the coiled retractable type.

3.4.2.5.4.3 Inverter type 28-volt system. A 12 to 28-volt inverter shall operate from the 12-volt battery and shall be rated for continuous duty.

3.4.2.5.5 Cable supports. Cable supports shall be furnished. Cable supports shall be located at the rear of the vehicle cab. Stowage for the cable shall be by means of a hook and hanging loop, or a protective holding bracket. When a hook or loop method is used, the cable shall be furnished with a hanging loop, so attached as to insure that the plug is pointed down when the jumper cable is stowed on the hook provided or on the pogo stick type hose tender. Protective bracket, when furnished, shall be mounted at the rear of the cab, below the roof line, and to the side of the cab rear window. The protective bracket shall hold the cable plug(s) within the bracket housing to preclude entrance of water into the electric cable plug terminals.

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3.4.2.5.6 Wiring schematic. A wiring schematic for the electrical system specified herein shall be furnished. The schematic shall be on a plate (see 3.1.1.1.3). The schematic shall depict the 12 volt lighting system and the 28-volt lighting system. Complete operating instructions shall be furnished. The schematic shall also include the electrical starting system.

3.4.2.5.7 Blackout lighting. When specified (see 6.2), 12-volt blackout lighting shall be furnished. Blackout lighting drawings will be provided when specified.

3.4.3 Gear train. Gear train shall meet or exceed the requirements specified in 3.3 through 3.3.1.6.

3.4.3.1 Transmission. An automatic transmission shall be provided. The transmission shall be capable of automatically upshifting and downshifting in at least the four upper gear ratios. With the detent in any fixed gear ratio, the transmission shall not seek any other gear ratio unless grade and load conditions increase with fixed engine power. The transmission shall have not less than five forward and one reverse gear ratio. Normal driving selector shall provide not less than four gear ratios without manual movement of the selector. Input torque, horsepower, and revolutions per minute and rated capacity shall be greater than that delivered by the vehicle engine. The transmission and components shall be capable of accepting the GVWR loads specified. The transmission shall include an oil screen, oil cooler and an oil filter. Transmission shift level sequence, starter interlock and transmission braking effect shall conform to FMVSS No. 102. Certify that the transmission is matched and rated for use in conjunction with interfacing components.

MARK AS FOLLOWS: Do not tow vehicle over 20 mph and no more than 15 miles.

3.4.3.2 Hydraulic torque converter. A hydraulic torque converter for the transmission shall be furnished. A red temperature light shall be furnished on the vehicle instrument panel to indicate when the converter oil is above manufacturer's recommendations. The stall torque ratio shall be the transmission manufacturer's recommendation to meet the performance specified in 3.3.

3.4.3.3 Power take-off. Power take-off opening(s) and heavy duty drive gears shall be furnished. The power take-off opening(s) shall conform to SAE J704. The openings shall be utilized for the hydraulic elevating fifth wheel. The power take-off drive rating shall provide a torque of not less than 300 pound foot and be rated for continuous duty. Power take-off controls shall be located in the vehicle cab in an accessible position to the driver , and shall be guarded to prevent accidental actuation.

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3.4.3.4 Neutral start switch. Neutral start switch for the transmission shall be furnished.

3.4.3.5 Speedometer drive outlet. A speedometer drive outlet for the transmission shall be furnished.

3.4.4 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, or torque developed through the maximum gear reduction. Drive line shall be balanced to preclude vibration in conformance with the manufacturer's recommendation. Means shall be furnished to lubricate all bearings and journals without removing the drive line from the vehicle. Certify that the drive line is matched and rated for use in conjunction with interfacing components.

3.4.5 Frame. Chassis frame shall be furnished. The frame shall be manufacturer's standard for the vehicle furnished. Frame shall have sufficient structural strength and resisting bending moment to at least equal the loads imposed with the fifth wheel in its lowered position and in its maximum elevated position under the specified operating conditions with the vehicle loaded to specified GVWR and GCWR. In addition, the frame and its components shall be capable of meeting the transportability requirements specified in 3.1.1.4.2 and 3.5.1. Crossmembers, as a minimum, shall be provided rearward of the transmission and at the rear of the chassis frame. Chassis frame rails shall be cut-off immediately to the rear of the axle bracket or the frame crossmember closest to the rear of this bracket. In any case, the rearmost back face of the frame member shall not extend more than two inches past the rearmost back face of the tire. Back face of each frame member shall be provided with a dock bumper. Bumpers shall be easily removed for service.

3.4.6 Suspension. Vehicle shall be furnished with a front suspension system with components having a rated capacity at least equal to the dynamic and static loads imposed on each member, measured at the ground, with the vehicle loaded to the specified GVWR. The rear axle shall be resiliently mounted to the vehicle. The axle resilient mounting shall have a rated capacity at least equal to the dynamic and static loads imposed on each member, measured at the ground, with the vehicle loaded to the specified GVWR. Sufficient clearance shall be furnished between the vehicle frame and the inner tires. Vehicle shall be furnished with hydraulic double acting shock absorbers at the front axle. Sufficient clearance between shock absorbers and front brake components shall be furnished, when the shock absorber is extended to manufacturer's extended length.

3.4.7 Axles. Front and rear axles shall be furnished. Axle ratings shall be at least equal to the dynamic and static loads imposed on each axle, measured at the ground, with the vehicle

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loaded to the specified GVWR. The front and rear axles shall be provided with oil lubricated wheel bearings and hub seals to resist contamination of the lubricants. Certify that the axles are matched and rated for use in conjunction with interfacing components.

3.4.7.1 Front axle. The front axle shall have a manufacturer's rating of not less than 12,000 pounds and shall be complete with springs, steering components and all other necessary parts. Camber and caster shall be alined to the manufacturer's recommendations. Front axle stops shall be set to meet the turning radius, and when the vehicle is making its hardiest left or right turn, and when any one wheel is elevated or below ground level, the vehicle tires shall have sufficient clearance to prevent the tire coming in contact with any vehicle component.

3.4.7.2 Rear axle. The rear axle shall be of the two-dual wheel type. The rear axle shall be a single speed, double reduction having a certified manufacturer's rated capacity at 35 mph of at least 33,000 pounds. The axle ratio shall be not less than 11.36:1.00. The axle and its components shall be capable of withstanding the stresses and loads imposed at maximum GCWR. Sufficient clearance shall be provided between the axle components and the fifth wheel and its components.

3.4.8 Wheels, rims, tire and tubes. Vehicle shall be equipped with single front and dual rear wheels. Rims, wheels and tire ratings shall conform to Tire and Rim Association recommendations at not more than 30 mph for the type and size of tires furnished. Tire, rim, and wheel sizes shall be of the same manufacturer on all wheels on each vehicle. Wheels shall be interchangeable between the front and the rear axles. Disc wheels shall be provided. Rims shall be the largest; preferred size. Side rings shall be compatible to the wheel rims. Wheels and rims shall conform to Federal Motor Safety Standard No. 120. Wheels shall have a rated capacity to at least equal the static and dynamic loads imposed. Fasteners shall be torqued to manufacturer's specification. Wheels and rims shall be treated, primed and finish painted as specified in 3.1.1.1. Certify that the wheels are matched and rated for use in conjunction with interfacing components.

3.4.8.1 Tires. Tires shall be tube type. Tread pattern shall provide traction on wet hard surface roads. Tires shall provide maximum resistance to heat build-up and capable of off-road operation. Side walls shall be cut, tear and crack resistant. Tires shall be at least 12.00 by 20-16 ply rating and shall have a rated capacity at least equal to the static and dynamic levels imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to specified GCWR. Tires shall be in accordance with Group 3 of ZZ-T-381M, under NSN 2610-00-489-8041 and Product Code 123-340-988 with tire flaps size R20-80-0. Tubes shall be NSN 2610-051-9454, Product Code 193-423-200. Tires shall be SRL-l tires for 8.50V rim. Tires shall conform to

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Federal Motor Safety Standard No. 119. Tires shall be of the continuous duty type. Tire capacity may be rated in accordance with the Service Load and Inflation Table, Load Limits at Various Speeds in Section 2 of the Tire and Rim Association Incorporated Year Book, at 30 miles per hour.

3.4.8.2 Inner tubes. Inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tire flaps shalervice Load and Inflation Table, Load Limits at Various Speeds in Section 2 of the Tire and Rim Association Incorporated Year Book, at 30 miles per hour.

3.4.8.2 Inner tubes. Inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tire flaps shall be provided for tube-type tires in accordance with The Tire and Rim Association recommendations. Valves shall be accessible for service from the roadside or curbside of the vehicle.

3.4.8.3 Spare wheel and carrier assembly. Spare wheel or rim and carrier shall be furnished and installed in a readily accessible location on the vehicle. Means shall be provided to secure spare wheel assembly to the carrier to prevent theft of the spare wheel and tire in transit. Cable or chain with nylon or canvas sleeve and a padlock shall be provided.

3.4.8.3.1 Spare tire. An inflated spare tire mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size, tread pattern and ply rating furnished on the vehicle.

3.4.9 Brakes. Brakes shall conform to the Federal Motor Carrier Safety Regulations 393.40 through 393.51, as specified herein. The brake system shall be capable of withstanding salt air and sea mist. Brakes shall be of a size recommended by the brake and axle manufacturer. Certify that the brake system is matched and rated for use in conjunction with interfacing components.

3.4.9.1 Service brakes. Vehicle shall be equipped with full air brakes. The anti-lock system depicted in Federal Motor Vehicle Safety Standard No. 121 shall not be furnished. The braking system, complete with all necessary components, shall include at least the following:

a. Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm) at 1250 rpm. Air hoses shall conform to SAE J1402. Manufacturer shall not use any exterior wire braid on air hoses that will be in contact with electrical components. Hoses shall be harnessed.
b. Air storage reservoir(s); each tank equipped with safety and check valves between compressor and last reservoir tank. Each tank equipped with drain valves.
c. Foot control, suspended or treadle type.
d. Limiting valve for front brakes, combined with quick release valve.
e. Quick release valve for rear-brakes.

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f. Air pressure gage, visible to the driver.
g. Low air pressure warning, visible and audible.
h. Alcohol aspirator with unbreakable transparent container, capable of accepting minus 50°F.
i. Automatic moisture ejector capable of operating at minus 50°F.
j. Air control valves.
k. Service brake stop lamp switch.
l. Air chambers.
m. Four-way relay emergency valve.

3.4.9.2 Trailer brake control system. In addition to components specified in 3.4.9.1, a trailer brake control system shall be furnished. Trailer brake control system shall at least include:
a. Identification of emergency and service lines. Emergency painted red and marked.
b. Coincident control of trailer brakes with prime mover foot control.
c. Independent hand control for trailer brakes.
d. Prime mover protection valve with dash control and automatic break-away.
e. Trailer stoplight operative with foot brake and with hand control for trailer brakes.
f. When pintle assembly is specified in 3.1.1.5, air connectors with cover for trailers with "gladhand" couplers mounted at the rear of the vehicle and located to preclude interference with the trailer. Dummy "gladhand" couplers with chains. Air connectors shall be in accordance with SAE J318. Emergency air connector painted red and marked.
g. Two connecting air hoses conforming to SAE J1402 and not less than 110 inches in length, equipped with coiled spring hose guards, and "gladhand" quick connector on trailer end of hoses located at rear of cab on a pogo stick type hose tender. Emergency coupler painted red and marked.
h. Spring supports on pogo stick type hose tender with dummy "gladhand" connectors to retain hoses when not in use. Hose tender shall not be mounted on cab roof.
i. See 3.1.1.5 for air lines forward of the rear crossmember.

3.4.9.3 Parking brake. A parking brake shall be furnished. The parking brake shall be capable of being manually applied by the operator seated in the driver's seat and shall be in accordance with Federal Motor Carrier Safety Regulation 393.41.

3.4.10 Cab. Manufacturer shall furnish a standard half-width tilt cab. Cab shall be mounted to preclude vibration. Body shall be insulated and weather stripped for inclement weather specified herein. A street side cab door, and curb side cab door, when provided, shall be

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provided with external key operated locks. Driver's seat side window shall be mechanically operated with a hand operated crank. The curb side window shall either be mechanically operated or shall be of the sliding type. When a sliding type window is furnished, it shall be provided with a means to preclude entry from cab exterior. Manually operated air vents shall be furnished near the floor level. Interior dome light with frosted lens and an off and on metal, heavy duty switch shall be provided. Open grating step(s) and grab handle(s) shall be furnished to enable personnel to ascend the vehicle for purpose of cleaning the windshield and to facilitate entry of personnel into the vehicle cab and onto the truck-tractor deck plate. The steps shall not interfere with personnel when cab is tilted for servicing the vehicle. Interior and exterior reflecting surfaces shall conform to Federal Motor Vehicle Safety Standard No.107. Cab roof shall be reinforced to preclude penetration of falling objects and endangering personnel when a 100 pound weight is dropped from a distance of 10 feet onto the center of the roof, at minus 50°F, personnel shall not be injured. Cab shall be capable of tilting by means of a hydraulic, manually operated pump. A lock shall be provided to prevent cab tilting rearward. Window insulated channels and weatherstripping shall be secured so as not to come loose.

3.4.10.1 Driver's seat. A driver's seat shall be furnished. The driver's seat and seat back shall be upholstered, and capable of horizontal and vertical adjustments of the seat assembly, seat and the seat back by the driver while in a seated position. The seat back shall be approximately 24 inches, measured from the seat cushion to the top of the seat back. The seat back shall be capable of tilting fore and aft. The seat cushion shall be capable of raising or being lowered. The seat assembly shall be of the mechanical type and not air operated due to climatic environments. Manufacturer's standard fastening and one pair of seat belts shall be furnished and installed. Seat belts shall be of the retractable type. Seating shall conform to Federal Motor Vehicle Safety Standard No. 207. Seat belt anchorages shall conform to Federal Motor Vehicle Safety Standards No. 209 and No. 210.

3.4.10.2 Door locks. Door locks and retention shall conform to FMVSS No. 206. Door locks, handles, latches and associated hardware shall be of the heavy duty type for highway truck application. Handles shall not break off when turning handles from ground level.

3.4.10.3 Window glazing. Window glazing material shall conform to FMVSS No. 205.

3.4.10.4 Flammability of interior materials. Flammability of interior materials shall conform to FMVSS No. 302.

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3.4.10.5 Rear window. A rear window shall be provided. The window and window operating mechanism shall be accessible to the driver in a seated position. The exterior of the window shall be provided with a screen. The screen shall be hinged and secured on the outside so as to enable maintenance of the window. The screen shall be of flattened, expanded metal, one-half inch opening and of 16 or 18 gage. The frame, screen and associated hardware shall be of corrosion resistant material.

3.4.10.6 Pocket for equipment record folder and manual. A pocket for the vehicle operator's manual and the Government Furnished Log Book Binder (NSN 7530-01-065-0166) shall be furnished in the vehicle cab. Equipment Control Record DD Form 2408-9 will be inserted in the Log Book Binder for each vehicle (see DD Form 1423).

3.4.10.7 Engine housing and cab aft guard plate. A 3/8 inch steel plate shall be furnished at the rear of the engine housing and the vehicle cab. The rear window shall not be obscured by the plate. The steel plate shall protect the housing, engine and the vehicle cab and driver from damage due to sliding loads on the towed vehicle and when coupling to a towed vehicle. Plate shall be bolted to the vehicle for removal of any vehicle component. Plate shall accept the acceleration factors of Federal Motor Carrier Safety Regulation 393.106.

3.4.10.8 Front bumper and bumper guard. Channel type heavy-duty front bumper and bumper guard as specified in 3.4.10.8.1 shall be furnished.

3.4.10.8.1 Front bumper guard. A front bumper guard shall be furnished. The bumper guard shall be hinged at the bottom and braced to allow the vehicle cab to be tilted forward. The guard shall at least be two-inch by three-inch steel square tubing haying a minimum wall thickness of 0.1875 inches, with equally spaced vertical uprights of two-inch square tubing having a minimum, wall thickness of 0.1875 inches. The height of the bumper guard shall protect the lower front of the engine compartment and the area below the cab windshield. Bumper guard corners shall be rounded to protect personnel.

3.4.11 Power steering. Power steering shall be furnished. Power steering mechanism shall enable vehicle to be manually steered in event of hydraulic steering or engine failure. An oil filtering system shall be furnished. Sheaves where used, shall be aligned and of the correct pitch diameter to preclude belt from run-off. Belts shall be of the correct length and width to preclude belt interference with any component. Steering column shall be braced and secured and properly alined. Steering control system shall conform to Federal Motor Vehicle Safety Standard

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No. 204. Turning radius shall not be more than 24 feet, curb to curb (see 3.4.7.1) as defined in paragraph 2.3 of SAE J645, 1979 Edition.

3.4.12 Windshield wipers and washers. Vehicle shall be equipped with windshield wiper and windshield washer. The windshield wiper(s) shall at least conform to the requirements of SAE J198, Federal Motor Carrier Safety Regulation 393.78 and Federal Motor Vehicle Safety Standards No. 102 and 104. Windshield wiper shall be multi-speed type and operated by an air motor.

3.4.13 Cab heater and defroster. Manufacturer's standard fresh air heater and defroster with controls shall be capable of providing adequate heat and window visibility in ambient temperature of minus 25°F in accordance with SAE J382.

3.4.13.1 Heater. Vehicle shall be provided with a hot water heater with blower and fresh air intakes and discharge outlets to the floor and outlets to assure a flow of heated air through the vehicle cab. Heater shall provide crew comfort of not less than 45°F when the outside ambient air temperature of -25°F. The heating system shall be provided with adjustable controls to provide outside air, recirculated air or a variable mixture as desired to circulate through the main heater core. In addition, the heater blower shall be capable of providing outside air without circulating through the heater core. Heater outlets shall be adjustable by the operator. Heater hoses shall not interfere when cab is tilted.

3.4.13.2 Defrosting system. Defrosting system shall be furnished. Defrosting system shall conform to Federal Motor Vehicle Safety Standard No. 103 and SAE J381 and 382 and at an ambient temperature of minus 25°F to provide 180 degree windshield visibility. Defrosting system shall be complete with blower and shall be capable of defrosting the windshield with heated air or with outside air. Controls shall be furnished. Defroster outlets shall be provided.

3.4.14 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator, to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to driver when seated in driving position. Control and illumination shall be in accordance with FMVSS No. 101. Controls and operating mechanism shall be guarded or positioned to preclude personnel from inadvertently operating the control or mechanism.

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3.4.15 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. Control, identification and illumination shall conform to Federal Motor Vehicle Safety Standard No. 101. The following minimum equipment shall be furnished:

a. Key-operated ignition switch. Keys for door(s) and ignition shall be keyed alike.
b. Ammeter and voltmeter.
c. Fuel gage.
d. Oil pressure gage with warning light.
e. Air pressure gage with warning light and audible system.
f. Engine temperature gage with warning light.
g. Speedometer in mph and in kilometers and with recording odometer in mph and in kilometers.
h. Tachometer. (may be combined with g).
i. Sunvisor.
j. Hydraulic fifth wheel elevating and descending controls.
k. Fifth wheel lock and unlock mechanism located inside the vehicle cab and accessible to the seated driver. Fifth wheel lock and unlock mechanism shall be provided with a guard or a means to prevent accidental actuation of these controls.
l. Power take-off control.
m. Front door mounted armrest on driver's side.
n. Driver's compartment ventilator other than window(s) by driver's feet.
o. Flood lamp switches.
p. Heat and oil gages or warning devices recommended by the transmission manufacturer.

3.4.16 Rearview mirrors. Outside rearview mirrors, each with supporting arms, shall be mounted on the street side of the cab and on the curb side of the vehicle. The curbside mirror shall be mounted on square tubing and braced to preclude vibration. Each mirror shall have not less than 80 square inches of reflective surface. Mirrors shall be capable of folding toward the vehicle cab. An inside rearview mirror shall be furnished. Mirrors shall conform to Federal Motor Vehicle Standard No. 111.

3.4.17 Horn. Manufacturer's standard electric and air horn shall be furnished.

3.4.18 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the instrument panel.

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3.4.19 Tool box. A tool box shall be provided. The tool box shall be mounted on the side of the vehicle. The tool box shall be 18 inches in height, 18 inches in width and not less than 5 inches in depth. A door with a continuous hinge and a hasp for a 1 1/2 inch padlock shall be furnished. The door shall be approximately 18 inches by 18 inches.

3.5 Hydraulic lift fifth wheel. Unless otherwise specified (see 6.2), a hydraulic lift fifth wheel shall be furnished. The hydraulic fifth wheel unit shall be power operated, all cast steel, heavy duty, with a rated capacity to lift and support an imposed load of not less than 50,000 pounds on the truck tractor fifth wheel assembly. The fifth wheel shall have a lifting range of not less than 16 inches, measured in a horizontal plane, from the top of the fifth wheel plate in its lowest position to the top of the plate in the fully raised position. When specified (see 3.5.5), the fifth wheel shall have a lifting range of 31 inches, from the lowest to the highest position (see table II.) Fifth wheel and its mounting structure shall have a design capacity to withstand a drawbar pull of 60,000 pounds without residual deformation to the mounting, fifth wheel or its vehicle structure. The fifth wheel shall be capable of being manually locked and unlocked in event of hydraulic and pneumatic failure. The fifth wheel shall be of the fore and aft rocking type with a 36 inch diameter cast steel coupler plate supplied with beveled approach forks. The cavity below the fifth wheel shall be provided with a water drain hole. Fifth wheel shall not interfere with the rear axle, differential or any hydraulic component. The inclined angle of the approach ramps shall be designed to accept the semitrailer when negotiating 14 degree ramps without any interference of the semitrailer undercarriage and the ramps. The top surface of the inclined angle of the approach ramps shall provide clearance above the vehicle tires, when coupling to a semitrailer, when the fifth wheel is tilted rearward to its maximum aft angle for coupling purposes. The approach ramps shall be of the box type, with top inside edges beveled, and the top surface of high tensile steel, to preclude kingpin being gouged or damaged. The vehicle fifth wheel assembly, its components and mounting, shall have sufficient structural integrity to accept the dynamic and static imposed loads when the towed vehicle has a center of gravity of 109 inches above ground level (see 3.2.9, Table II, NOTE). The fifth wheel unit shall conform to Federal Motor Carrier Safety Regulations 393.70 a, b, e and g; and 393.71. Mode of mounting the fifth wheel shall be equal or similar to SAE SP-354.

3.5.1 Fifth wheel subframe. The fifth wheel assembly shall be provided with an inclined ramp presenting a continuous slope for semitrailer approach. When a sub-frame is provided, it shall be mounted to the vehicle chassis frame rails. Means shall be provided to preclude

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deformation of the sub-frame and the vehicle chassis frame rails. The vehicle chassis frame and fifth wheel sub-frame assembly and its mounting shall be capable of accepting the loads imposed during coupling, uncoupling, ascending and descending ramps up to 14 degrees from horizontal and the effects of weight transfer up and down the specified grades and inclines and have sufficient structural strength that when a 20 foot, 20 ton container is dropped at the front of the 40 foot semitrailer, 6 inches, that the truck tractor structural member will accept the moments imposed. Calculations shall be furnished to the procuring activity.

3.5.2 Fifth wheel hydraulic system. The fifth wheel unit shall include a hydraulic system and mechanism arranged to elevate the fifth wheel; shall incorporate a hydraulic or pneumatic system to open and close the semi-automatic lock for SAE J700 kingpin; shall include a device to prevent false locking in the lock; shall embrace remote control to permit all fifth wheel operations, including operation of the power take-off from within the vehicle cab; and shall include all components, such as power take-off, pump, hydraulic reservoir and suction and return filters to remove contaminants (filters shall be accessible for service), and hose and fittings, to comprise a complete installation. Means shall be provided to prevent payload descending in event of hydraulic failure, and a manual means provided to lock and unlock the fifth wheel assembly and to bleed down the elevated fifth wheel. When bleeding down the fifth wheel hydraulic system, the hydraulic fluid shall return to the reservoir. The pneumatic system, if furnished, shall be provided with an alcohol aspirator with an unbreakable transparent container and capable of operation at minus 50°F. Fluid shall not leak out the vent of the reservoir when the fifth wheel is down and with payload. Instruction plate shall be provided.

3.5.3 Clearance. The clearance from the centerline of the kingpin to the pogo stick type hose tender when furnished, with the fifth wheel at all elevated positions, shall be not less than 70 inches and not more than 98 inches. The landing wheel clearance from the center line of the kingpin to a vertical plane on the outside edge of the rear tires, with the fifth wheel in the lowest position, shall be not more than 57 inches, and the chassis frame rails shall be not more than 30 inches. Under no condition shall the kingpin be positioned to the rear of the centerline of the axle. The unladen level height from the ground to the top of coupler plate of the installed fifth wheel unit with the unit in its lowest position shall be 48 inches plus 1 inch minus 0 inch. Actuation through the full lifting range shall not cause the center of the kingpin lock to shift more than 10 inches, measured along the longitudinal centerline of the vehicle (see Table II).

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3.5.4 Grating plate. A steel walkway grating plate shall be furnished. The plate shall cover the area forward of the hydraulic fifth wheel assembly; width of the frame and the fuel tank. The plate shall be secured with corrosion resistant hand operated toggle clamps to facilitate maintenance of the fuel tank(s) and areas below the plate. The plate shall be located as close to the back of the vehicle cab as possible and extend rearward to the maximum length practicable. When a step fuel tank is provided, the top of the tank inward of the tank outlet fittings that are not protected, shall be provided with removable guards to preclude damage to tank fittings. Guards may be secured if fittings are accessible for service. Provisions to allow access of personnel climbing onto the plate from the sides of the vehicle cab shall be provided. Access holes with grommets, located for maintenance of fittings and other components shall be furnished. The plate shall be of sufficient structural strength for the use intended. The plate toggle clamps shall be bolted, and not welded, to any chassis frame member.

3.5.5 Elevating hydraulic fifth wheel. When specified (see 6.2), an elevating hydraulic fifth wheel shall be furnished. The fifth wheel shall be capable of elevating from down position to elevating position at least 31 inches, and from ground level at least 79 inches. The elevating fifth wheel shall meet the requirements specified in 3.5, 3.5.1, 3.5.2 and 3.5.3.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government Inspector, contractor shall service and adjust each vehicle for operational use including at least the following: Focusing of lights, adjustment of the engine, electrical 12 and 28 volt system, and power plant heaters; and brake system; filling and charging of battery; alinement and balancing of the front wheels; inflation of all tires; cycling the fifth wheel assembly through its entire range of elevations; complete lubrication of chassis, engine and running gear, fifth wheel plate and the kingpin activating mechanism with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol and water in equal parts by volume; servicing of windshield washer reservoir with water and appropriate additives; visual inspection to verify that no accessory or component will interfere with any other accessory or component under dynamic conditions; and that all fasteners are in place and torqued for their respective size.

3.7 Safety. All equipment, or exposed portions of the equipment which are subject to extreme temperatures; rotating or reciprocating parts which are of such a nature or so lubricated as to become a hazard to operating personnel, shall be insulated, fully enclosed or properly guarded.

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3.8 Keys. All locks on the vehicle, except for the spare wheel assembly, and all vehicles on the contract shall have identical keys. Locks are to be keyed alike. Two keys for each vehicle shall be furnished.

3.9 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Initial production test vehicle road test and examination. The initial production test vehicle will be examined and road tested GCWR by the contractor to assure that the vehicle operated in accordance with the contract requirements. The test vehicle will be operated a minimum of three (3) continuous hours during which it will be subjected to fifteen (15) cycles of pick-up, coupling, delivery of not less than one half (1/2) mile maneuver and park in a limited parking area, uncoupling and dropoff.

4.2.1 Vehicle weight. Initial production test vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, the operator weight at 200 pounds, and the payload to provide the specified GVWR. Calculated imposed loads on front and rear axle will be used to evaluate the capability of the suspension, axles and tires furnished to meet performance requirements.

4.2.2 Initial production test vehicle road test. One initial production test vehicle will be examined and road tested in accordance with 4.2. Upon successful completion of road test as specified in 4.2, this vehicle will be subjected to the following initial production test examinations and conditions.

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4.2.2.1 Highway operation test. The initial production test vehicle will be examined and road tested GCWR to assure that the vehicle operates in accordance with the contract requirements. The test vehicle shall be highway operated at a minimum speed of 25 mph for at least 1000 miles in accordance with 4.2.2.3 and shall maintain full governed speed for at least three miles during each two hour road test session.

4.2.2.2 Maneuverability and placement. The test vehicle shall be operated in accordance with 4.2.2.3 and figure 1 in two identical yard areas with improved surfaces, and level within 2° to assure satisfactory operation and maneuverability. Yard areas shall be limited by pylons, stanchions or chalk marks to a 115 foot maximum width, and during maneuvering and placement no part of tractor or semitrailer shall cross over these designated borders. The vehicle shall be used in conjunction with a 40 foot semitrailer loaded to provide GCWR.

4.2.2.3 Test procedures. The test vehicle will be operated on a daily basis in the following manner. Run the vehicle for two hours in accordance with highway operation test procedures of 4.2.2.1, follow immediately with maneuverability and placement test of 4.2.2.2 for a two hour period. Continue to alternate the two tests for the duration of an eight hour day. Testing will continue in this manner until the vehicle has accumulated 1000 miles.

a. Maneuverability and placement test will begin by coupling the yard tractor to a loaded trailer which has been preparked between stanchions and/or pylons which are situated to represent dimensionally two 40 foot semitrailers separated by 12 feet. After coupling, fifth wheel of tractor shall be raised to full height. Fifth wheel will then be adjusted to a normal operating height determined by the driver. The semitrailer shall be maneuvered out of its parking position without any part of tractor or semitrailer encroaching upon those adjacent areas which have been set up to represent the two other semitrailers.

b. The semitrailer shall then be delivered a distance of 1/2 mile to a yard area identical to the first and shall be backed into its parking area so that it is centered, and even with the other represented semitrailers. The semitrailer shall then be uncoupled and the tractor fifth wheel fully lowered. The tractor will then pull forward a minimum of 20 feet, reverse, and re-couple with the semitrailer; again raising the fifth wheel fully.

c. The semitrailer will then be delivered to the original area, and the above procedure repeated; this will complete one full cycle. Five cycles will be completed in one hour.

d. One yard area may be used in lieu of two, provided 1/2 mile distance is traveled, and at least five cycles are completed in one hour. The trailer must be backed into the parking area by alternating left hand and right turns to demonstrate turning ability in both directions.

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4.2.2.4 Schedule of examinations.

Initial Production	Defect	Requirement Paragraph
X	101. Cooling system not as specified.	3.1.1.3
X	102. Coolant system and indicators not as specified.	3.1.1.3.1
X	103. Towing, tiedown and lifting devices not as specified.	3.1.1.4, 3.1.1.4.1
X	104. Splash shields not as specified.	3.1.1.9
X	105. Carbon monoxide and NO_X concentrations not as specified.	3.1.1.11

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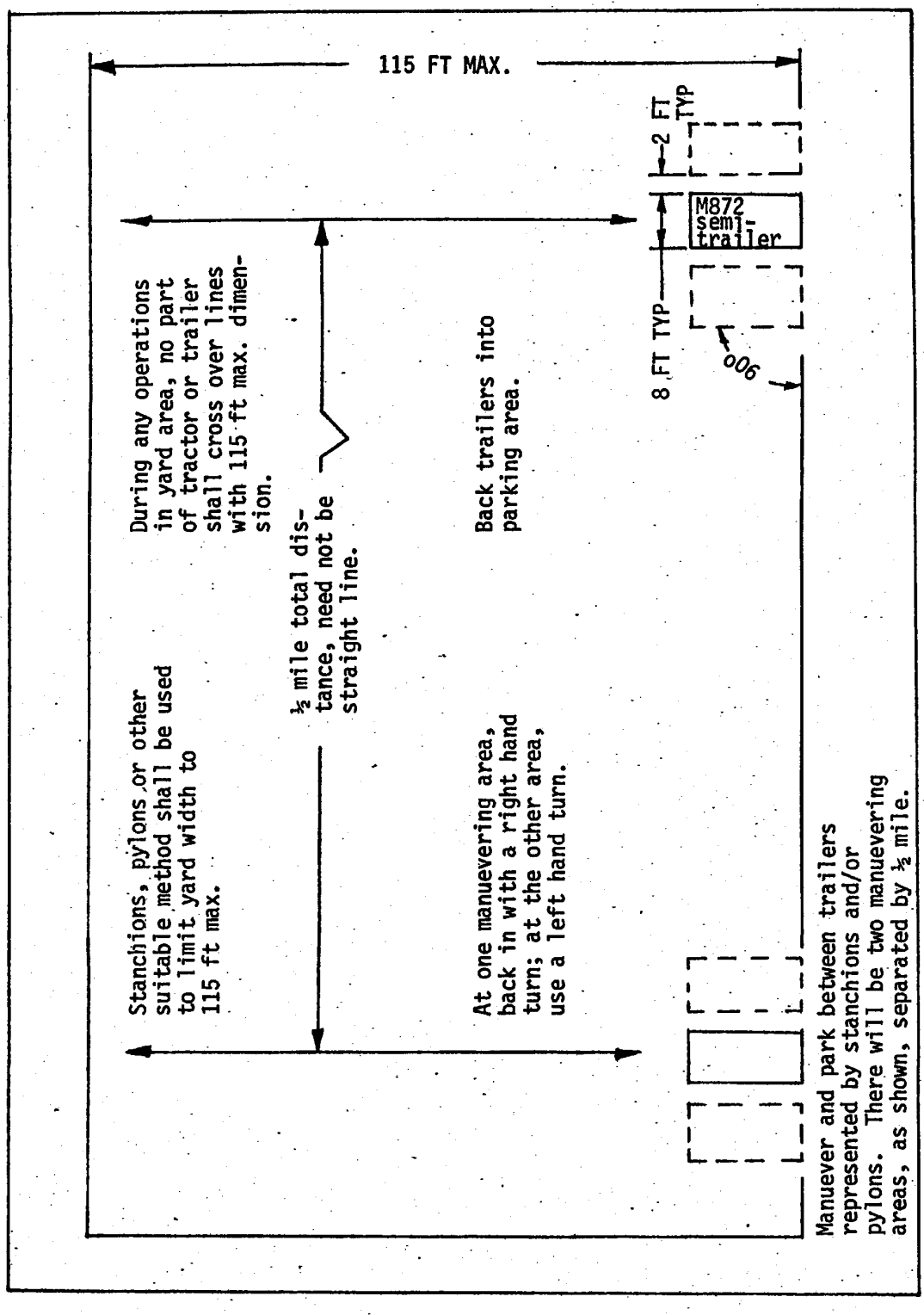


FIGURE 1. Manueverability and placement.

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X	106. Sound level not as specified.	3.2.3, 3.2.3.1, 3.2.3.2
X	107. Vehicle not capable of operation in climatic conditions.	3.2.4.1
X	108. Gradeability not as specified.	3.3.1.1
X	109. Low speed not as specified.	3.3.1.2
X	110. Reverse speed not as specified.	3.3.1.3
X	111. Maximum geared speed not as specified.	3.3.1.4
X	112. Inching capabilities not as specified.	3.3.1.5
X	113. Vehicle incapable of sustained operation as specified.	3.3.1.6
X	114. Service brakes not as specified.	3.3.2, 3.4.9.1
X	115. Parking brake not as specified.	3.3.2.1, 3.4.9.3
X	116. Engine governor not as specified.	3.4.1.3.5
X	117. Engine starting equipment not as specified.	3.4.1.3.6
X	118. Electrical wiring not as specified.	3.4.2
X	119. Transmission not as specified.	3.4.3.1
X	120. Trailer brake control system not as specified.	3.4.9.2
X	121. Power steering not as specified.	3.4.11
X	122. Heater not as specified.	3.4.13.1
X	123. Fifth wheel hydraulic system not as specified.	3.5.2
X	124. Clearance not as specified.	3.5.3

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4.2.2.5 Test conditions. Unless otherwise specified in 4.2.5, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Prior to commencement of initial production testing, vehicle shall have been prepared for operation in accordance with 3.6. During test, the vehicle will receive servicing and adjusting at intervals specified in the contractor's literature previously furnished to the Government representative.

4.2.2.6 Driver-operator. Driver shall be an experienced operator and shall be capable of, and required to give objective opinions of the vehicles' handling, operational, and human engineering characteristics to the Government representative. These opinions shall be evaluated by the Government and contractor with regard to importance and necessary action.

4.2.2.7 Rejection. Permanent deformation, overheating, malfunction, component interference, failure to meet performance requirements, failure to complete any test, or leakage of fuel, lubricants or hydraulic fluid will be basis for rejection of the initial production test vehicle. If such deficiencies do arise, they will be corrected on site and the initial production vehicle test will be repeated. Acceptance of the initial production test vehicle shall not constitute a waiver by the Government of its rights under provisions of the contract.

4.2.3 Radio frequency suppression verification. The vehicle manufacturer shall furnish, with the questionnaire (vehicle descriptive data), a description of all interference suppression devices used to suppress the vehicle in accordance with SAE J551. Data shall be sufficient for visual determination that all required suppression devices are installed on the vehicle. When specified in 3.1.1.6, the EMI shall be tested to verify that the components meet the requirements of MIL-STD-461. Certified test reports shall be provided that the vehicle meets or exceeds the requirements of 3.1.1.6 when specified or the requirements of 3.1.1.6.1. Method and visual verification shall be provided.

4.2.4 Fifth wheel lift capacity verification. The vehicle manufacturer shall furnish certified test data verifying the capability of the hydraulic lift fifth wheel to lift and support a load of 50,000 pounds over the kingpin throughout the lifting range as specified in 3.5.

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4.2.4.1 Fifth wheel (coupling interference). The truck-tractor incline ramps when coupling to a semitrailer, shall be of sufficient incline to prevent the front of the semitrailer contacting the truck tractor tires. Demonstration shall be performed, when the truck tractor tires are at maximum pressure recommended by the Tire and Rim Association for the size and ply rating of the tires (see 3.5).

4.2.5 Arctic provisions. When specified in 3.1.1.10, arctic provisions shall be examined and operated for not less than 50 miles at 20 to 30 mph at rated GVW to verify conformance to contract.

4.2.6 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. Contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.2.7 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.3 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicles conform to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 has been accomplished.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of the vehicle from the manufacturer's plant to the initial receiving activity, shall be in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for use by the US Army for shuttling and spotting semitrailers at seaport terminals, truck terminals, and depots.

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6.2 Ordering data. Purchaser should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number and date of this specification.
b. Identification of appropriate service for painting and marking, if other than required (see 3.1.1.1.1).
c. Concealed marking, if required (see 3.1.1.1.1(b)).
d. Vehicles supplied to foreign nations, if required (see 3.1.1.1.1(c)).
e. Rear mounted pintle assembly, if required (see 3.1.1.5.1).
f. Electromagnetic interference (EMI) suppression, if required (see 3.1.1.6).
g. Arctic provisions, if required (see 3.1.1.10).
h. Blackout lights, if required (see 3.4.2.5.7).
i. Fifth wheel, 16 inch elevating, if not required (see 3.5).
j. Fifth wheel elevating 31 inches, if required (see 3.5.5).

6.2.1 Data requirements. Data requirements incorporated in DD Form 1423 shall be as specified on Data Item Description Form DD Form 1664 and delivered in accordance with the approved contract data requirements list DD Form 1423. Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for all speed requirements for the vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item should include vehicle model number, wheel base, engine model number, transmission model number and rear axle model and vehicle type. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1. Tire factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

Table 2. Altitude factor. 1.00

Table 3. Rolling factor. 0.501

Table 4. Area factor. 0.261.

Table 5. Velocity factor. 16.0

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Table 6. Altitude factor. 1.00

Table 7. Chassis friction horsepower. Use applicable power unit GVW (to nearest, higher 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for the geared speed. For GVWR and engine speed beyond the range of this table, factors shall be extrapolated.

Table 8. Grade factor. 1.87

Table 8A. Correction factor. Not required.

Table 9. Road factor. 0.1

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